DESIGN MILESTONES

Alara Bayrak S046039

PROJECT OVERVIEW

Portable Electric Filter Coffee Brewing Device

Design description:
A portable electric filter coffee machine for office workers who love coffee but lack the time to prepare it at home by focusing on workflow interruptions and dependence on shared office equipment.The device consists of four modules: a coffee storage compartment, heating chamber, filtering chamber, and thermos body; connected through threaded screw connections. Water is heated in the upper chamber and released into the filter, where the coffee is brewed before collecting in the thermos below. There is a button that starts the brewing process and afterwards pours the water to the filtering chamber after its boiled as well as a mechanism that stops the coffee from dripping when the thermos is removed. The mechanism works by pushing up a piece and creates an opening for the coffee to enter the thermos when its closed and the opening closes up after the thermos removed.

Keywords:
- Workflow continuity
- Portable brewing
- Modular design
- Office coffee ritual
- Personal workspace autonomy
- Filter coffee
- Compact heating system
- Car-compatible device

I will design a portable electric beverage preparation device that allows office workers to prepare hygienic, quiet and high quality drinks at their desks without relying on shared equipment.

Office workers struggle with hygiene and spatial control in shared office environments because beverage preparation often depends on bulky and shared systems leading to disturbed work hours or personal time and comfort.

I will develop a portable electric beverage preparation device designed for desk and in-car use, enabling office workers to prepare drinks independently while maintaining workflow continuity and reducing their reliance on shared office equipment.

I will develop a portable electric filter coffee machine for office workers who love coffee but lack the time to prepare it at home by focusing on workflow interruptions and dependence on shared office equipment.

95°
85°
85°
85°

REFLECTIONS

While completing this design challenge AI definitely made me realize that I wasn't considering enough of the details that I was supposed to be considering and this project certainly helped me realize that AI isn't as incompetent as I thought it was because I've been trying to use it for the wrong reasons this whole time. While considering what kind of kettles to design I had my ideas but I just couldn't imagine what they would look like but when I asked Chat GPT to generate images for me it helped my ideas properly fall into place.

AI didn't significantly change my core ideas of designing a compact, desk friendly hot beverage preparing device however it helped me think more critically about the small details. Through the discussion I began to think harder about the safety and stability of my design as well as the noise control, cable management and for it to be easily cleanable and have multiple temperature options. Therefore, AI mostly expanded my thinking rather than changing the direction I was going. Ai also made me realize that the design I was thinking of is already available on the market so I'd either have to think of something else or find something that would make my design different.

AI helped me narrow down the tension ideas that I was thinking of and helped me understand the parts of the assignment that I didn't understand which made a significant difference in my thinking. It also helped me understand whether my design direction was possible with todays technological advancement or not.

AI, rather than changing my thinking, helped me organize my thoughts and my research findings into clearer categories and sharpened the connections between user pain points and opportunity ideas. It also helped me narrow down the target user group more precisely instead of keeping it way too broad. AI also helped me understand the parts of the assignment I wasn't sure about.

During this stage, AI helped me question whether my concept was actually solving the tensions I selected or if I was just adding new features. It made me think harder about the relationship between workflow interruption and autonomy instead of only focusing on the portability of the device. It also helped me recognize when an idea might complicate the project instead of strengthening it. I was thinking about adding an app integration but AI said that it would complicate the design so I steered away from the idea.

I was struggling to understand if what I was imagining was possible but after speaking with Chat GPT I realized that there might be a chance that it might be possible and I continued on with my idea.

DECISIONS

I accepted the idea of focusing on ergonomics and safety and refining the idea rather than focusing on the technicalities since I was just getting started on the project. I also rejected the image it created for one of the ideas because the previous one it generated was too small but when I asked it to generate it again it took away some of the key features so I had to ask it to generate it two more times.

I accepted the suggestions related to safety improvements, steam control, and noise reduction because they directly strengthen my concept but I chose not to accept more complex smart features such as app integration or to make it highly technical, as they would shift the project away from its main focus of being simple and desk integrated. My design direction will prioritize practical usability rather than overcomplicating the product.

I accepted some of the formats it gave me because I wasn’t sure about how I was supposed to answer some of the questions I was given but I rejected the wording I was given and fully used my own wordings for everything. I also rejected AI’s suggestion to simplify my project by not making it multi-functional because I thought it would take away from my design direction.

I accepted the structured formats for insights and the refined problem statement because they helped clarify what I was thinking and the design direction I was going for. However, I rejected the wording that it used and adjusted everything according to my interviews and observations.

I rejected the suggestions that were contradicting my tensions or generally changing my core design idea too much. I rejected suggestions that shifted the focus too much towards simplification or removing multifunctionality, since I believe the temperature options, stirring and self-cleaning support my core design idea by reducing interruption and dependency.

I was originally going to make the whole body metal but when I sent my drawing and idea to AI it kept on generating it in glass and after ı thought about it I decided to change that part to glass because it seemed like a feature that the user would be satisfied with.

Reflection Summary

AI helped me expand my concepts and my ideas while also help provide the technical details of my project. After finishing this project I've realized that AI truly does cut the time projects take for short by helping decide on the technicalities that would have taken longer otherwise.

Design Decision Summary

I accepted a lot of the ideas provided for me while also making sure that I was the one in control by giving detailed prompts and providing my own ideas. AI was generally helpful besides the times where it would just send me the same responses over and over again and generate the same pictures. AI also sometimes provided the wrong technical details in which case I just had to test it on my own.

AI Role Tags

- Clarifier → Helped explain design tensions
- Re-framer → Suggested different concept directions
- Expander → Generated alternative mechanisms
- Critique → Evaluated my technicalities of the heating and brewing system

SELECTED IMPACTFUL PROMPTS

How can a portable coffee preparation device prevent dripping when the thermos is removed?
AI role: Expander
AI Response summary
Suggested valve systems and anti-drip mechanisms.

What internal mechanism would allow heated water to move into the filter chamber?
AI role: Clarifier
AI Response summary
Explained heating plate placement and valve release mechanism.

What dimensions should a portable coffee device have to fit a car cupholder?
AI role: Expander
AI Response summary
Provided realistic dimensions and modular proportions.

What materials and thickness are appropriate for a portable heated beverage device?
AI Role: Critique
AI Response Summary:
Recommended stainless steel walls (~1 mm) and a thicker heating plate (1.5–2 mm).

WHAT WAS REJECTED AND ACCEPTED

Multiple temperature options: I initially wanted to make a device that would allow for coffee, tea and instant noodles to be made with different temperature options for each one. However, after considering all the factors I realized that the idea would complicate my target audience as well as make my device too complicated. Therefore I decided to design a device solely for filter coffee.

Cross section valve. After thinking about all solutions to my dripping problem I thought about using cross slit valves to solve my problem but after I observed different coffee machines and saw the solutions that manifacturers came up with, I decided to use a mechanism that pushes up a piece and creates an opening for the coffee to enter the thermos when its closed and closes up after its removed.

Bag clip: When I first started the project I wanted to make a portable device that would clip onto bags for easy use but later on I decided that it was unnecesary and wouldn’t provide much of a solution to anything.

Mug concept: When I was considering the size of the device I thought about deigning a portable device that would be in the shape of a mug but later on decided that it would make the mechanisms harder and it would not fit in a car cup holder.